

SIMATS ENGINEERING

CO2-ASSESSMENT TOOL - 2

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ETHICAL HACKING

Question 1 – theHarvester

Open Terminal and run:

```
sudo apt update
theHarvester --version
```

If it says "command not found":

```
sudo apt install theharvester
```

Run the OSINT scan:

```
theHarvester -d tesla.com -b all
```

To save the report:

```
theHarvester -d tesla.com -b all -f tesla_report
```

Record the publicly available results under the following categories:

- Email addresses
- Subdomains
- Hosts
- IP addresses

Categorize discovered email addresses as Support, Sales, Human Resources, Administration, or Technical Teams based on the apparent role of each address.

Risk Analysis: Public email addresses, subdomains, hosts, and IP information can help an attacker understand an organization's structure and prepare believable spear-phishing, credential-harvesting, or impersonation pretexts. No interaction with discovered accounts should be performed.

Countermeasures: Reduce unnecessary public email exposure; configure SPF, DKIM and DMARC; use MFA; deploy email and URL filtering; conduct awareness training; monitor public information exposure; and remove obsolete DNS records.

[INSERT YOUR THEHARVESTER TERMINAL SCREENSHOT HERE]

Question 2 – GoPhish

If GoPhish is already installed in your lab:

```
cd gophish  
sudo ./gophish
```

Open your browser:

```
https://127.0.0.1:3333
```

Log in and perform the authorized phishing-awareness simulation using only the dummy employee accounts provided in the lab.

Create and take screenshots of:

- Dashboard
- Dummy Users / Groups
- Email Template
- Landing Page
- Campaign
- Results – Delivery Rate, Open Rate and Click Rate

Safe Email Template Example:

```
Subject: Security Awareness Training Test  
Hello {{.FirstName}},  
This is an authorized cybersecurity awareness simulation.  
Please use the training link as instructed during the lab session.  
Regards, Security Awareness Team
```

Result Formulas:

```
Delivery Rate = Delivered Emails / Total Emails × 100  
Open Rate = Opened Emails / Delivered Emails × 100  
Click Rate = Clicked Links / Delivered Emails × 100
```

Recommendations: Train users to verify senders and links, conduct regular authorized simulations, use MFA, deploy email filtering, configure SPF/DKIM/DMARC, and provide a simple method for reporting suspicious emails.

[INSERT YOUR GOPHISH DASHBOARD / CAMPAIGN RESULT SCREENSHOT HERE]

Question 3 – Nmap

Check version:

```
nmap --version
```

Run the aggressive scan:

```
nmap -A scanme.nmap.org
```

Save the report:

```
nmap -A scanme.nmap.org -oN scanme_report.txt
```

From the output, record:

- Target IP address
- Open ports
- Detected services
- Service versions
- Operating system details
- Traceroute information

Social Engineering Analysis: Information about operating systems, services, and software versions can make a false technical-support story sound more believable. An attacker could use accurate technical terminology to impersonate an employee or technician and pressure IT support into revealing information or bypassing normal procedures.

Defensive Recommendations: Verify user identity before password resets or account changes, avoid disclosing internal technical details, minimize exposed services, patch systems, use MFA, monitor suspicious authentication activity, and train help-desk staff against social engineering.

[INSERT YOUR NMAP -A TERMINAL SCREENSHOT HERE]